

7.2 Performance Evaluation Metrics

Q: Confusion Metrics

	Actual positive	Actual Negative
predicted positive	85	4
predicted Negative	2	9

(a) Accuracy

(b) Error Rate

(c) sensitivity

(d) specificity

(e) Precision

(f) Recall

(g) F-score

Solution

(a)

85	4
2	9

TP	FP
FN	TN

$$acc = \frac{TP + TN}{TP + FN + FP + TN} = \frac{85 + 9}{85 + 4 + 2 + 9} = 94\%$$

TP	FP
FN	TN

(b)

$$er = \frac{FN + FP}{TP + FN + FP + TN} = 6\%$$

TP	FP
FN	TN

(c)

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{85}{85 + 2} = 97.7\%$$

TP	FP
FN	TN

(d)

$$\text{specificity} = \frac{TN}{TN + FP} = \frac{9}{4 + 9} = 69.2\%$$

TP	FP
FN	TN

(e)

$$\text{precision} = \frac{TP}{TP + FP} = \frac{85}{85 + 4} = 95.5\%$$

TP	FP
FN	TN

(f)

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{85}{85 + 2} = 97.7\%$$

TP	FP
FN	TN

(g)

$$F = \frac{2 \times \text{precision} \times \text{Recall}}{\text{precision} + \text{Recall}} = \frac{2 \times 95.5\% \times 97.7\%}{95.5\% + 97.7\%} = 96.6\%$$

TP	FP
FN	TN